

TACTICS

Four-Team Force-vs-Force Range

Collaborator implementation roadmap

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Working direction. This is a proposed technical handoff, not an adopted stack or purchase authorization.

The system we need

Build a reproducible, isolated range in which **four teams of up to four students can participate at the same time**. Each team receives its own enclave, identity namespace, network boundary, infrastructure state, and evidence namespace. A separate practice enclave lets students rehearse the same skills outside a live event without reaching any team enclave.

The range exists to carry skills from existing Techniques material into coordinated Tactics work. A service belongs in the first release only after curriculum and scenario collaborators link it to actual Techniques lessons and established work-role material. Mail using Postfix and Dovecot is the proposed pilot profile; it is not final until that mapping passes. Detailed Techniques content remains at jboyce1.github.io/ppsCTF until a migration actually occurs.

Each team enclave must provide:

- functioning services and synthetic data appropriate to the scenario;
- a controlled path for student administration and assigned offensive missions;
- observable service health, student actions, mission outcomes, and recovery actions;
- a reliable reset and independent rebuild from versioned definitions; and
- isolation from range management, protected evidence, backups, the internet, the practice enclave, and teams that are not assigned opponents.

Infrastructure as code must reproduce the four team enclaves from the same versioned profile while generating unique environment IDs, VM IDs, addresses, identities, secrets, state, and evidence records.

Student exercise phases

Student phase	Duration / required outcome
1 • Hardening	2–4 hours. Each team establishes a functional service baseline, inspects and hardens assigned systems, and records changes while required workflows continue to operate. No cross-team mission access is open.
2 • Force-on-force	2–4 hours. All four teams execute assigned missions and defend their own enclaves at the same time. Only paths named in the event mission matrix are open. Service continuity, mission outcomes, capture gaps, and recovery actions are recorded.
3 • AAR reporting	2–4 hours. Cross-team access is closed, active attack sessions are terminated, evidence is preserved, and each team produces an evidence-supported after-action review.

The phases may run as separate sessions. Preserve event state and evidence between sessions. The complete exercise is 6–12 hours plus setup and reset.

Four-team interaction model

Four simultaneous teams do not require unrestricted four-way access. The first release uses an explicit mission matrix. The recommended starting arrangement is two matched force-on-force pairs:

- Team A is paired with Team B.
- Team C is paired with Team D.

This gives all four teams simultaneous offensive and defensive work while limiting routing, troubleshooting, and scoring complexity. Later scenarios may use a ring or another declared assignment, but no team receives any-to-any access by default. The event manifest identifies attacker sources, target services, allowed ports, start time, stop time, and mission IDs for every opened path.

How to use this handoff

1. Verify hardware, dedication, and backups in section 1 before changing anything.
2. Build the minimum four-team product using the stack and gates in sections 2 and 3.
3. Implement the shared technical contracts in sections 4 through 7.
4. Accept the release only after a full four-team rehearsal, using section 8.

Example addresses, VLAN IDs, and software choices are implementation defaults. They require overlap checks and collaborator validation before use.

1 | Hardware and implementation boundaries

Reported resources — verify before configuration

Resource	Proposed role and required verification
Dell PowerEdge	Run Proxmox and the four team enclaves, the practice enclave when capacity permits, and staff support VMs. Verify exact model, CPU, RAM, usable storage, storage health, NICs, and iDRAC. Earlier records conflict between R730 and R750; neither label is assumed.
Protectli appliance	Proposed range boundary: VLAN gateways, default-deny inter-zone policy, phase rules, controlled egress, and collaborator VPN if remote access is required. Verify model, installed firewall OS, ports, throughput, current use, and console access before repurposing or reinstalling it.
Cisco switch	Carry approved VLANs between the Protectli and the PowerEdge as Layer 2 transport. Verify model, module, IOS, licensing, optics, port capacity, and whether the switch is dedicated to the range.
Backup destination	Hold protected configuration exports, VM backups, release manifests, and evidence archives independently of the systems being backed up. Availability, capacity, access control, and restore performance must be verified before configuration work.
Student and collaborator access	Confirm up to 16 simultaneous student seats, staff and operator sessions, and remote collaborator needs. Decide whether the first event is on-site, remote, or hybrid before treating WireGuard or a jump host as a release dependency.

First physical deliverable

Before any reconfiguration:

1. Record each device, physical port, cable, switch port, VLAN, gateway, management path, and responsible collaborator.
2. Determine whether the Protectli or Cisco switch currently serves another network. If either is not free to repurpose, stop and open a named remediation or migration task.
3. Export current configurations into protected independent storage.
4. Record checksums and prove that another collaborator can retrieve the exports.
5. Identify a console path and a rollback trigger for every network or platform change.

Four-team capacity starting point

For the proposed mail pilot, begin capacity planning with two target VMs per team:

Guest type	Per guest starting allocation	Four-team total
Mail server	2 vCPU / 2 GB / 24 GB disk	8 vCPU / 8 GB / 96 GB disk
Student client VM	2 vCPU / 2 GB / 24 GB disk	8 vCPU / 8 GB / 96 GB disk
Team target subtotal	4 vCPU / 4 GB / 48 GB disk	16 vCPU / 16 GB / 192 GB disk

Also budget:

- one automation and deployment VM;
- one evidence and scoring VM, or separately sized collector and scorer;
- optional jump and session-capture capacity if remote access is required;
- one practice mail server and client if they will run at the same time as the event;
- host memory and storage headroom;
- template, backup, evidence, and rebuild storage; and
- logging growth for four concurrent teams across all three phases.

Budget support VMs at 4–8 GB RAM each, then measure. These figures are planning inputs, not verified requirements. CPU may be overcommitted cautiously. RAM, usable storage, backup capacity, disk latency, and log growth require direct measurement.

At the hardware gate, run a go/no-go review. If four team enclaves plus required support services do not fit with safe headroom, change the VM profile, move support services, or schedule the practice enclave separately. Do not place self-hosted GitLab on the event host if doing so reduces four-team reliability.

Ownership boundary

Staff and authorized collaborators control the physical network, Proxmox, deployment state, secret references, collectors, scoring, backups, and phase transitions. Students receive guest-level access within their assigned enclave. Vulnerable services remain inside the range and use synthetic mail, users, and files. No target mounts infrastructure backups, hypervisor storage, source-control state, or protected evidence archives.

2 | Minimum first-release stack

The minimum first release is **four matching team enclaves**, plus one independently resettable practice enclave that may run at a different time if capacity is constrained.

Build path: private Git repository of reviewed definitions; staff automation VM running pinned Terraform and Ansible; PowerEdge/Proxmox versioned Linux template; five isolated environments below. Network path: named student sessions on assigned clients; Protectli zone and phase policy; Cisco VLAN transport; host and firewall logs to a protected collector; service probes to an event, mission, and recovery ledger; four-team evidence export and human AAR review.

Environment	Contents	Assigned opponent
Team A	Mail server and student client	Team B
Team B	Mail server and student client	Team A
Team C	Mail server and student client	Team D
Team D	Mail server and student client	Team C
Practice	Mail server and student client	None. Never routes to Teams A–D.

Component	Minimum responsibility
Private Git, Terraform, and cloud-init	Version the scenario and infrastructure. Clone a prepared template with unique VM IDs, addresses, and bootstrap identities. Keep separate state for Teams A–D and practice. A private repository may be hosted externally. Self-hosted GitLab is not required for the minimum.
Ansible and the selected service	Configure local name resolution, the selected service, identities, logging, and synthetic fixtures. If mail passes the Techniques gate, use Postfix for authenticated submission and Dovecot for retrieval. Reuse the same roles across all team and practice enclaves.
Logs, audit, and functional probes	Collect authentication, selected audit, firewall, and service events. Test actual service behavior. Record mission and recovery decisions using shared event, team, and incident IDs.
Named access and reporting	On-site named accounts or SSH may satisfy the first release. If collaborators or students require remote access, build and validate a jump and session-capture endpoint before WireGuard becomes an event dependency.

Packer, protected GitLab deployment runners, Guacamole, SIEM indexing, Zeek/Suricata, and AI-assisted feedback remain expansion work. Self-hosted GitLab may proceed in parallel only after network, backup, access, and capacity validation. It must not block image, IaC, service, or evidence work.

3 | Minimum build sequence and release gates

Each build step starts from accepted outputs of its dependencies. Accepted means a different collaborator reproduced or witnessed the pass condition, and the downstream receiver confirmed the result is usable.

Step	Build next	Pass condition
M0 • Controls and service choice	One task-ID scheme, staffed validation roles, private repository, secret-handling rules, change and rollback practice, and a Techniques-to-service mapping decision.	Every claimable task has one ID. At least two people can independently validate. Protected source exists. The proposed first service is linked to real Techniques material, or it is replaced before service build begins.
M1 • Inventory and preservation	Verify hardware, dedication, port map, backup destination, configuration exports, capacity, and recovery paths.	Equipment identity is known. Switch and firewall are available for range use, or a named remediation exists. Protected exports are retrieved and checksum-verified. Hardware can plausibly support four teams, or a documented capacity change is made.
M2 • Network and platform	Install or validate the firewall OS and Proxmox. Isolate management. Configure Layer 2 VLAN transport, gateways, default deny, platform bridges, backup, and restore.	Tagged test guests for Teams A–D and practice reach their gateways. They cannot reach management, another closed enclave, undeclared internet, or undeclared IPv6. One guest backup restores successfully.
M3 • Reproducible empty enclaves	Prepare and version one Linux cloud-init template. Implement the allocation registry, Terraform clone module, separate state, and guarded lifecycle commands.	Create and destroy four empty team enclaves twice. VM IDs, addresses, identities, state, and evidence namespaces are unique. Resetting one leaves the other three unchanged.
M4 • Service and evidence	Configure the selected service in practice first. Deploy time, log collection, functional probes, actor and session mapping, heartbeat and gap detection, and evidence export. Add only the declared firewall exceptions for services that now exist.	Practice service succeeds. Expected misuse fails after hardening. Configuration rerun is idempotent. A named action, service transaction, failed probe, recovery, and logging gap appear in a checksummed export.
M5 • Four-team scenario and operations	Clone matching A–D enclaves. Install event-specific mission proofs. Build practice hardening, attack, and recovery activities. Implement versioned phase-rule sets and runbooks.	Each assigned pair can execute its mission from the real entry point. Fixes preserve required service. Practice reset affects no team. Closing Phase 2 terminates existing cross-team connections.
M6 • Integration and release	Rehearse all phases with four teams at planned concurrency. Export four AAR packages. Restore evidence and rebuild one team from the release manifest.	At least two hours per phase. All four teams operate concurrently. Resource peaks remain within safe limits. Isolation, scoring, capture-gap detection, phase transitions, restore, and independent rebuild pass.

Working rules

- Do not claim network isolation because nothing is connected or listening. Prove it with tagged guests after the switch, firewall, and Proxmox path exist.

- Do not create guessed DNS, NTP, logging, or scoring exceptions before those endpoints exist. Establish default deny at M2, then add named exceptions and rerun the communication matrix at M4.
- Do not require GitLab migration or GitLab backup before images, Terraform, or the evidence collector can progress.
- Do not require WireGuard for on-site identity mapping. Require it only when the documented participation model needs remote access.
- Draft event runbooks early. Validate them against the real four-team environment at M5 and M6.

4 | Network and phase-control contract

Example allocation — verify overlaps before use

Zone	Example configuration contract
Management	VLAN 10 / 10.77.10.0/24. Named staff and collaborators only. No student or target route.
Automation	VLAN 20 / 10.77.20.0/24. The deployment system receives scoped hypervisor and configuration access.
Evidence	VLAN 40 / 10.77.40.0/24. Example collector 10.77.40.10, scorer 10.77.40.20. Allocate actual endpoints before installing exceptions.
Team A	VLAN 111 / 10.81.1.0/24. Protectli gateway 10.81.1.1.
Team B	VLAN 121 / 10.82.1.0/24. Protectli gateway 10.82.1.1.
Practice	VLAN 131 / 10.83.1.0/24. Protectli gateway 10.83.1.1.
Team C	VLAN 141 / 10.84.1.0/24. Protectli gateway 10.84.1.1.
Team D	VLAN 151 / 10.85.1.0/24. Protectli gateway 10.85.1.1.

These are examples, not assigned production values. Compare them with school, home, VPN, management, and collaborator networks. Maintain a versioned allocation registry that rejects duplicate environment IDs, VM IDs, VLANs, and active subnets.

Routing principles

- Use the Cisco as Layer 2 transport for range VLANs. Do not create team-routing SVIs that bypass the Protectli.
- Use a VLAN-aware Proxmox range bridge with no management IP. Keep hypervisor administration on the management path.
- Students cannot edit VM NIC tags.
- Use host firewalls where same-VLAN traffic would not cross the Protectli.
- Default deny between zones. Student enclaves cannot reach device management, backups, source or state stores, protected evidence, practice, or closed team enclaves.
- Block arbitrary internet egress. Block IPv6 until an equivalent policy is designed and tested.
- Every permitted path must identify source, destination, protocol and port, purpose, and lifecycle.

Baseline service exceptions

After endpoints exist and have addresses, permit only declared flows: for example, target DNS and NTP to designated endpoints, TLS syslog to the collector, and scorer-initiated functional checks. Validate directionality. Permission for a target to send logs does not grant it broad access to the evidence zone.

Phase 2 mission paths

The release manifest contains a mission matrix. For the recommended first event:

Attacker	Target	Allowed service	All other teams
Team A assigned client(s)	Team B declared target	Mission ports only	Blocked
Team B assigned client(s)	Team A declared target	Mission ports only	Blocked
Team C assigned client(s)	Team D declared target	Mission ports only	Blocked
Team D assigned client(s)	Team C declared target	Mission ports only	Blocked

If mail is selected, authenticated submission on TCP 587 and IMAPS on TCP 993 are the likely functional paths. SMTP 25 is opened only if a mission explicitly requires server-to-server SMTP. At Phase 2 closure, disable the mission rules, clear matching states, and prove that an already established cross-team session is terminated. Keep cross-team paths closed during hardening, AAR, and practice.

5 | Provisioning and state contract

Each environment definition must declare its identity and source versions. A four-team event therefore has distinct records for team-a, team-b, team-c, and team-d, plus practice.

```
event_id: event-2027-01-example
environment_id: pilot-c
team_id: team-c
profile: mail
mode: competition
image_version: <pinned-id>
scenario_version: <git-commit>
network:
  vlan: 141
  cidr: 10.84.1.0/24
checks:
  - mail_roundtrip
opponent_assignment: team-d
seed_reference: <staff-secret-reference>
```

Terraform emits unique VM IDs and private IPs as Ansible inventory. Cloud-init sets unique hostnames and bootstrap identities. Ansible configures the selected service, lab trust, synthetic names such as team-c.test, logging, and fixtures. Generate secrets per deployment outside Git. Remove or scope bootstrap access after configuration.

Maintain separate state for each team and practice environment. A failure, plan, apply, reset, or state lock in one environment must not overwrite another. Keep platform state separate from enclave state and treat state as sensitive.

The staff-run interface should eventually expose:

- validate
- plan ENVIRONMENT

- deploy ENVIRONMENT
- verify ENVIRONMENT
- phase EVENT PHASE
- export-evidence EVENT
- reset ENVIRONMENT

Validate all targets against the allocation registry. Refuse unknown environments. Reset only the named enclave after confirming event closure, successful evidence export, and operator intent.

6 | Service, identity, and evidence contract

Service choice gate

Before implementation, curriculum and scenario collaborators must identify the Techniques lessons and existing work-role skills reused by the proposed profile. If mail cannot be grounded in actual material, either add a Techniques learning path before the event or select a first service that already has one. Do not invent or rename a work role to make the mapping appear complete.

Proposed mail profile if the gate passes

Use separate synthetic domains such as team-a.test, team-b.test, team-c.test, team-d.test, and a practice domain. Configure authenticated submission and encrypted retrieval using a lab CA distributed to clients and probes. Deny external relay and delivery. The functional probe submits a uniquely identified synthetic message, retrieves the same message, verifies content, records latency and result, and cleans up probe messages.

Identity and event records

Assign individual student guest identities and named access sessions. Do not share a team-wide administrative identity when individual attribution is required. Keep VPN collaborator peers, event student accounts, synthetic service accounts, probe accounts, and staff deployment identities as separate identity classes.

Use common fields across all four teams:

```
event_id, environment_id, team_id, actor_id, session_id
source_id, timestamp_utc, receipt_timestamp_utc, event_type, evidence_id
service_id, result, latency_ms
mission_id, incident_id, opponent_team_id, reviewer_id
```

Forward authentication, selected audit, firewall, and service records to a staff-controlled collector using authenticated transport. Buffer locally during interruptions and emit a heartbeat. A stopped heartbeat after two expected intervals creates a capture-gap record. Use controlled time synchronization and flag drift.

Avoid blanket keystroke or secret capture. Document the selected audit scope and session-recording behavior in the participant notice. Target compromise can falsify future target logs, so corroborate important results with staff-controlled probes, firewall records, and named session records.

Mission validation and recovery

Use an event-specific synthetic proof per mission and team assignment. Store expected proof material in staff-only storage. Record attempted, submitted, and validated outcomes separately. Reject proof replay from another event, team, or opponent pairing.

Track service state as healthy, then failing, then recovering, then recovered. Open an incident on the first failed functional check. Confirm it on the next failure while retaining the first-failure time. Close only after two passing checks plus verification of the corrective action. Report availability, mission outcomes, recovery durations, capture gaps, and AAR quality separately.

7 | Practice and event operation

Practice product

The practice environment uses the same image, service roles, checks, mission validator, and recovery logic as the team enclaves, but it has its own network, state, identity, evidence namespace, quota, and reset policy. It must never route to Teams A–D.

The first release should contain three practice activities if the selected service supports them:

1. **Hardening:** identify and correct a designated weakness while the functional check continues to pass.
2. **Attack:** execute the designated synthetic mission inside practice and validate an event-specific proof.
3. **Recovery:** restore a deliberately disrupted service and satisfy the same two-pass recovery check used in competition.

These may reuse the event mission and recovery design. They should not become entirely new scenarios.

Operating a four-team event

Phase 1 — Hardening

Record event, image, scenario, configuration, and environment versions for all four teams. Start logging, heartbeats, and baseline probes. Open only own-team administration. Each team submits a change log. Exit only when the service baseline works, evidence receipt is verified, and unresolved failures are recorded.

Phase 2 — Force-on-force

Enable only the paths in the reviewed mission matrix. Confirm A-to-B and C-to-D reachability from the actual assigned attacker clients while all unassigned cross-team paths remain blocked. Start the common scored window. At the deadline, disable mission rules, clear relevant firewall states, verify active cross-team sessions are terminated, and export final service and event state.

Phase 3 — AAR

Provide each team read-only evidence for its own activity and the evidence needed to assess assigned opponent interactions. Protect other teams' identity mappings and unrelated data. Use isolated review copies if findings must be reproduced. Each team submits a timeline, mission evidence, recovery verification, remaining issues, and Techniques skills to practice next. A coach validates citations before reset.

AAR package

Export:

- one shared UTC event timeline;
- team-scoped actor and session references;
- service-check ledger for Teams A–D;
- mission assignment and validation records;
- recovery incidents and changes;
- configuration and release versions;

- capture-gap and staff-caused exclusion records; and
- a checksum manifest.

Confirm retention, access, and participant-notice rules before collecting student sessions. Keep the mapping between real identities and actor IDs restricted.

8 | Release acceptance and expansion

Four-team release acceptance

A collaborator release must attach or reference:

- release commit and image IDs;
- the four-team environment manifest and allocation registry;
- positive service tests for all four teams;
- negative isolation tests covering management, every unassigned team path, practice, protected evidence, backups, internet, and IPv6;
- successful assigned-opponent missions for both pairs;
- a repair that preserves required service operation;
- phase closure terminating active A-to-B and C-to-D test connections;
- logging-gap detection;
- access revocation when remote access is in scope;
- independent reset or rebuild of one team without changing the other three;
- evidence-archive restore and checksum verification; and
- peak CPU, RAM, storage, disk latency, network throughput, and evidence growth during the full rehearsal.

The release passes only after a full four-team rehearsal of at least two hours per phase. A two-team test cannot establish four-team capacity or concurrency.

Parallel self-hosted GitLab track

After network, backup, access, and capacity validation, collaborators may deploy self-hosted GitLab, protected deployment runners, and release jobs. GitLab should provide reviewed merge requests, protected source, issue tracking, CI, and eventually scoped deployment. It remains outside the minimum critical path until its resource use does not threaten four simultaneous teams, its backup and restore are proven, administration is reachable only by the declared secure path, and a private external repository remains available as fallback.

When GitLab CI is adopted, serialize apply, Ansible, and verify per environment, and use state locking. Do not split one deployment into jobs that can interleave against the same environment. Keep deployment credentials in protected jobs on an isolated runner.

Later expansion

After the four-team minimum is reliable:

1. Automate images with Packer.
2. Add DNS, web, NAS, or other service profiles one at a time after Techniques mapping.
3. Add Guacamole and assigned access VMs if browser-based remote participation is required.
4. Expand the practice pool and self-service lifecycle.

5. Add indexed evidence and tested network sensors.
6. Rehearse multi-service, multi-enclave operation at measured concurrency.

Optional AI assistance may draft feedback from indexed, redacted evidence only after the manual evidence and review path works. Require citations and human review.

First collaborator handoff

Staff the following functions before opening implementation work: project coordination, network, platform, remote access, source and release, services, evidence and scoring, scenario and curriculum, event operations, and at least two people able to perform independent validation.

Use one primary task ID per claimable unit of work. Large work packages should contain smaller child tasks, each ending in one observable validation event. Preserve the five-part definition of done:

1. prerequisites are accepted;
2. work is performed in the intended environment;
3. validation evidence is attached without secrets or student data;
4. someone other than the implementer reproduces or witnesses the pass condition; and
5. the receiving collaborators confirm that the output enables their downstream work.

Status of this draft

Draft 0.3 is the collaborator handoff for a four-team first launch. It keeps the original architecture — isolated enclaves, Techniques-to-Tactics reuse, three student phases, default deny, infrastructure as code, and evidence with capture-gap detection — and sets the first release at four simultaneous team enclaves using two declared opponent pairs.

Hardware values, software versions, network allocations, curriculum mappings, remote-access requirements, and event dates require validation before implementation. Task cards will be written after this minimum launch description is accepted.